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Evaluation of Land-Cover Model Predictions Through Citizen Science-Informed Reference Labeling in Fremont, California

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Abstract: Global land-cover products have made it easier to analyze landscapes around the world, but their use of dominant thematic classes can simplify more heterogeneous suburban landscapes. In this study, the European Space Agency's WorldCover 2020 product was evaluated within a 3 km × 3 km area of interest in Fremont, California. A total of 3,600 manually interpreted Map User reference labels across 36 Primary Sampling Units were created using high-resolution imagery and GLOBE Observer field photographs. These labels were then crosswalked to WorldCover classes and paired with WorldCover classifications at the corresponding Secondary Sampling Unit coordinates. At the SSU level, WorldCover had an overall thematic agreement of 65.2% (2,346 of 3,600), performing best when classifying Permanent water bodies and Built-up. WorldCover also classified a larger proportion of SSUs as Built-up and a smaller proportion as Grassland and Shrubland compared with the Map User labels. At the PSU level, dominant land-cover classes agreed in 29 of 36 PSUs (80.6%). As a result, WorldCover generally captured larger-scale land-cover patterns even when substantial differences remained at the SSU level. These findings demonstrate how nested comparisons can identify where global land-cover products represent broad suburban patterns well and where higher-resolution, locally informed interpretation is still needed. Since the reference labels were manually interpreted and crosswalked between legends, these results measure thematic agreement rather than absolute classification accuracy.

Keywords: land cover, remote sensing, citizen science

1. Introduction

Over the last half-century, remote sensing data has transformed our understanding of changing Earth systems. Satellite missions like the National Aeronautics and Space Administration's Landsat and the European Space Agency's Sentinel have been used extensively to track data about our planet, such as water storage, climate patterns, and even landscape evolution. Specifically on landscape change tracking, remote sensing has proven to be a powerful resource for observing how land cover (LC) and land use (LU) change around the world over time periods. As such, resources like the United States Forest Service's (USFS) Landscape Change Monitoring System (LCMS) (U.S. Department of Agriculture, Forest Service, n.d.) heavily rely on this remote sensing data to not only provide it in visual form for those seeking it, but they also utilize this remote sensing data to train models which serve to predict the landscape composition of any given area on Earth. As such, the use of high-quality data is incredibly important so that models are more accurate, studies using the data have smaller margins of error, and policies based on the data can be more targeted at issues in a changing landscape.

While the original satellites used for remote sensing data returned extremely low quality data, by today's standards, extensive progress has been made in trying to create new satellites for remote sensing missions, with the ability to return higher quality data through utilizing more advanced equipment. For

example, the latest in the Landsat series of satellites, Landsat 9, carries an Operational Land Imager 2 for 30 meter resolution (U.S. Geological Survey, n.d.) as opposed to the older Multispectral Scanner System (MSS) to improve image quality and provide additional spectral bands, which are very helpful for classification purposes. However, even with these upgrades over the past few decades, there still remains temporal and spatial resolution issues with the data returned back by these satellites. This can be attributed to several reasons, including that there simply aren't enough satellites orbiting Earth to take more frequent pictures of a certain place, or even that the sensors (including advanced digital cameras) simply do not have high enough resolution. Specifically focusing on the latter, issues with spatial resolution then invites misclassifications of landscapes from entities like LCMS, including overgeneralization of a region's LC/LU, or just completely incorrect classifications in general.

This study compared ESA WorldCover 2020 classifications with manually interpreted Map User reference labels at 3,600 SSUs nested within 36 PSUs in Fremont, California. The analysis examined (1) thematic agreement at the SSU level, (2) agreement between dominant PSU classes, and (3) differences in class composition across the AOI and selected PSUs. Because the reference labels were visually interpreted and harmonized between different legends, the results are treated as measures of thematic agreement rather than absolute product accuracy.

2. Study Area and Methods

2.1. Study Area



Figure 1. ESA WorldCover 2020 land-cover classes within the Fremont study area. © ESA WorldCover project 2020 / Contains modified Copernicus Sentinel data (2020) processed by the ESA WorldCover consortium.

The study was conducted on a 3 km x 3 km grid of Fremont, CA as the area of interest (AOI), as seen in Figure 1. Fremont is a suburban city which sits at the edge of Silicon Valley and has a history of largely being agricultural land while under Spanish rule and early into its incorporation into the US (City of Fremont, n.d.-c). As such, for a long time, Fremont's agricultural land use made it so that much of its land cover was not built-up, or just generally not built-up. This would change during the advent of the digital age, in which the population sharply increased due to widespread demand to move into Silicon Valley and the regions surrounding it (City of Fremont, n.d.-c). For Fremont specifically, it has become a manufacturing heart for Silicon Valley (City of Fremont, n.d.-d). As a result, the city has undergone much land development over the past forty years, mainly in an effort to scale residential, commercial, and

industrial land use to the growing demand over the past few decades.

Today, the city is not experiencing as much population growth, instead facing very minor, albeit present, population declines. Even so, Fremont's population has generally been estimated to be between 220,000 and 230,000 people throughout the years, so the population has more so plateaued than sustained a long term decrease (U.S. Census Bureau, n.d.). Looking at demographic information, the median age in the city was 39.5, in 2024 (U.S. Census Bureau, 2024), which is in-line with Fremont's population being primarily composed of families. Thus, it makes sense that much land in Fremont is currently zoned for residential purposes. To be specific, about 31% of Fremont's land area is residential properties and about 9% is covered by industrial areas. What is interesting about this is that despite a history of rapid land development, Fremont still reserves about 48% of its land area to be for open space, like parks and natural reserves (PlanIT Geo, LLC, 2020, p. 9). This is in-line with trends from the rest of the Bay Area, with regional policies emphasizing the conservation of natural land cover, such as tree cover (Metropolitan Transportation Commission, 2026; PlanIT Geo, LLC, 2020, p. 7).

Currently, the rapid development seen earlier in the city's history is not being replicated in the present day. However, even with very marginal population decline, Fremont is still experiencing increased land development zoning as the city targets increasing Fremont's presence on the map. In the recent past, this development has accelerated most starting in 2012, where the city finalized the master plan for zoning the Downtown Fremont project (City of Fremont, n.d.-a). This has expanded since then in the form of several policies, including the Fremont-Decoto Plan to further develop certain areas of the city (City of Fremont, n.d.-b). This is part of the wider Housing-Element plan that the City of Fremont has set forth, which lays out an 8-year plan to develop Fremont even further, between 2023 and 2031 (City of Fremont, 2023), matching the general trend of increased development through the modern day.

2.2. Data

This study utilized multiple sources of data, including citizen science observations of land cover, manually-classified units of LC, remote sensing data, aerial imagery, and satellite imagery from various satellite missions to document and contrast land cover classifications for Fremont, CA.

2.2.1. GLOBE Observer Data

The Global Learning and Observation to Benefit the Environment (GLOBE) Observer (GO) Land Cover tool was used to collect field observations used to support the manual reference data labeling process. For the AOI, we utilized the Adopt-a-Pixel 3 km (AaP3km), as proposed by Low et al. (2021) to create 36 primary sampling units (PSUs) in a 6 x 6 grid, positioned based on an AOI center point, to gather LC data from across Fremont. More information about this PSU set up is provided in Section 2.3. Each PSU's center was 500 m away from the center of any adjacent PSU, and composed of a 10 x 10 grid of secondary sampling units (SSUs). At each PSU, a total of 6 photos were taken: North, East, South, West, Up, and Down (Kohl et al., 2021; GLOBE Observer, n.d.), when possible. An example of the four cardinal direction photos can be seen in Figure 2. Land cover make-up was also documented at these sites by listing what percentage of the surrounding land cover was of each class. Collecting this data allowed for getting a better idea of what land cover looked like at close scales, since land cover elements such as shrubs and trees were more distinct on-the-ground than through satellite imagery. As such, this collected data represented References which could be used as a reference to compare against for remote sensing data.



Figure 2. Example of the four cardinal photos taken at each location. Up/down photos are not included. Photo credit: Aditya Kaul.

Not all PSUs were accessed due to geolocation errors, being located in inaccessible areas (such as private property), and timeline constraints. These limitations are covered more in depth later in Section 4.3.

2.2.2. ESA WorldCover Data

Remote sensing data was primarily sourced from Sentinel-1 and Sentinel-2 through the European Space Agency's (ESA) WorldCover (WC) project. WC was selected because its 10 m pixels provide finer spatial detail than LCMS's 30-m pixels (U.S. Department of Agriculture, Forest Service, n.d.; Zanaga et al., 2021; European Space Agency, 2021). Spatial resolution can affect classification performance, although finer resolution alone does not guarantee higher accuracy (Li et al., 2023).

From WC, LC classification data was collected for each of the 3,600 SSUs. This data was classified using WC's LC classes, which were: "Tree cover", "Grassland", "Shrubland", "Bare/sparse vegetation", "Permanent water bodies", "Built-up", "Cropland", and "Herbaceous Wetland".

2.2.3. Manual Classification Data

For each of the 3,600 SSUs in the AOI, I manually classified the LC found in that SSU using the LC classes provided by the USFS' LCMS (U.S. Department of Agriculture, Forest Service, n.d.). Labeling was done by using GO observations, RGB imagery from the United States Geological Survey (USGS) and United States Department of Agriculture's (USDA) National Agriculture Imagery Program (NAIP) for 2024 (California Department of Technology, 2024), and satellite imagery from Esri Landsat Explorer (ELE) (Esri, n.d.). An example of the labeled PSUs can be seen in Figure 3.

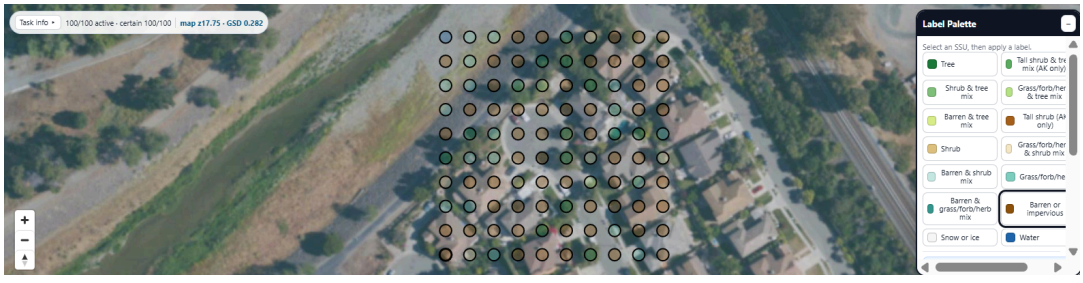


Figure 3. User interface for the labeling tool while on PSU 36, showing 100 SSUs over 2024 NAIP RGB aerial imagery (California Department of Technology, 2024).

2.3. Methods

As mentioned previously, to collect the data mentioned in Section 2.2.1, the AaP3km methodology was utilized in this study. The location of the center for each of the 36 PSUs was calculated by using a central point at coordinates 37.556042, –122.003245. From there, the positions of the 100 SSUs per PSU were also calculated. Navigation to each of the PSUs was done by generating a Comma-Separated Values (CSV) file with the coordinates of all the SSUs and exporting it to Google Maps. At each PSU, GO observations were taken at SSUs closest to the PSU's center; limitations to this will be covered later in Section 4.3

As mentioned previously for the manually labeled LC data, the classifications were done by using NAIP and ELE LC images along with GO observations. Through NAIP, I was able to gain access to land cover images from across the AOI at resolutions of 1 m or better. Then, taking the 36 PSUs, a grid was constructed where each of the 3600 SSUs were overlaid on the NAIP map by geographic position (latitude and longitude). Then, using the LC classifications provided by the USFS's LCMS, each SSU was classified based on the underlying NAIP imagery and the collected GO observations. The classifications used included: "barren grass mix", "barren impervious", "barren shrub mix", "grass forb herb", "grass shrub mix", "grass tree mix", "shrub", "shrub tree mix", "tree", and "water".

Original LCMS Label	Crosswalked WC Labels
Barren grass mix	Bare/sparse vegetation
Barren impervious	Built-up
Barren shrub mix	Bare/sparse vegetation
Grass, forb, herb	Grassland
Grass shrub mix	Shrubland
Grass tree mix	Tree cover
Shrub	Shrubland
Shrub tree mix	Tree cover
Tree	Tree cover
Water	Permanent water body

Table 1. Crosswalk used to harmonize the original Map User labels with ESA WorldCover classes.

When there was no access to a GO observation at a given PSU, SSUs were labeled according to NAIP imagery and cross-referenced with Landsat satellite imagery of the Fremont AOI. The images were accessed through Esri Landsat Explorer and renderers like "Natural Color", "NDVI Colorized" (where NDVI stands for "Normalized Difference Vegetation Index"), and "Urban" were used for the cross-referencing in the classification process.

Given that the manual classifications came from LCMS classes and I was comparing to data which used WC's classes, the manually classified data had to be crosswalked to allow for proper data analysis and comparison. Table 1 shows how each LCMS class was crosswalked to WC.

Each crosswalked WC label was chosen based on how much they thematically agreed with the corresponding LCMS label. For example, the class “Barren impervious” was assigned exclusively to land cover which was built-up, and hence it crosswalked to “Built-up”. A source of error in this study could arise from this label crosswalking, however, since LCMS labels capture secondary LC classes. As such, a label like “Grass shrub mix” gets reduced to simply “Shrubland” by the crosswalk, for all labels. This means that not only does this label crosswalk lose information about mixed SSUs, but converting those mixed labels into one or the other means that some labels end up incorrect. Going back to the “Grass shrub mix” example, if two SSUs are labeled as such, but one is more grass-dominant and the other is more shrub-dominant, no matter which WC class is used for the crosswalk, one of the SSUs ends up labeled incorrectly. Despite this, the selected crosswalk enabled direct comparison between the legends but remained a source of uncertainty.

3. Results

With the crosswalked data, I evaluated thematic agreement between the WorldCover model classifications and the Map User reference data for the AOI. From here on, these are referred to as the “model data” and “reference data,” respectively. Agreement was examined through the comparisons described below.

3.1. Total Label Count

The first comparison summarized total label counts by class, as shown in Figure 4. This was done by tallying the number of SSUs assigned to each class in the model and reference datasets. These totals show differences in overall class composition but do not measure whether individual SSU labels matched.

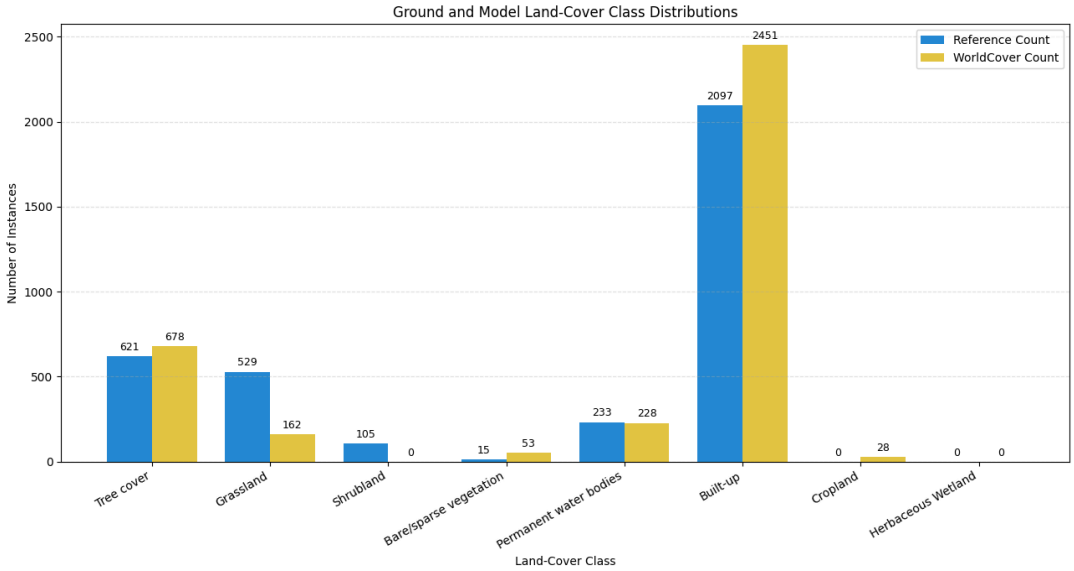


Figure 4. Class totals for the Map User reference and ESA WorldCover 2020 across 3,600 SSUs.

3.2. Confusion Matrix

The WorldCover classifications were compared with the manually classified reference data using a confusion matrix, as seen in Figure 5. Overall accuracy, user’s accuracy (UA), and producer’s accuracy (PA) were calculated following standard remote sensing accuracy-assessment practices (Olofsson et al., 2014). F1 scores were then calculated as the harmonic mean of UA and PA. These metrics can be found in Table 2.

	Tree cover	Grassland	Shrubland	Bare/sparse vegetation	Permanent water bodies	Built-up	Cropland	Herbaceous Wetland
Tree cover	273	12	0	11	8	315	2	0
Grassland	83	112	0	10	0	301	23	0
Shrubland	17	3	0	1	3	79	2	0
Bare/sparse vegetation	0	6	0	0	0	9	0	0
Permanent water bodies	11	0	0	2	217	3	0	0
Built-up	294	29	0	29	0	1744	1	0
Cropland	0	0	0	0	0	0	0	0
Herbaceous Wetland	0	0	0	0	0	0	0	0

Figure 5. Confusion matrix comparing harmonized Map User reference interpretations with ESA WorldCover 2020 classifications across 3,600 SSUs. Rows represent Map User classes, and columns represent WorldCover classes.

Land Cover Type	User Accuracy (%)	Producer's accuracy (%)	F1 score	User map reference count
Tree cover	40.2655	43.9614	0.420323	621
Grassland	69.1358	21.1720	0.324168	529
Shrubland	N/A	0.0000	0.0000	105
Bare/sparse vegetation	0.0000	0.0000	0.0000	15
Permanent water bodies	95.1754	93.1330	0.941432	233
Built-up	71.1546	83.1664	0.766931	2097
Cropland	0.0000	N/A	0.0000	0
Herbaceous wetland	N/A	N/A	N/A	0

Table 2. Class-specific UA, PA, F1 score, and Map User reference count for the ESA WorldCover 2020 comparison across 3,600 SSUs. NA indicates that the metric was undefined because the corresponding reference or WorldCover class count was zero.

3.3. Per PSU Metrics

The final comparison examined agreement between the primary and secondary labels assigned by the WC and reference datasets at the PSU level. This was found by grouping the 3,600 SSUs into 36 PSUs, and then calculating the most and second-most represented LC labels per PSU. The secondary classification was only considered if that class made up at least 10% of the SSU labels within the PSU to prevent areas from receiving a secondary LC classification which misrepresented that region. If two classes tied for primary label, one was assigned as primary and the other was assigned as secondary based on visual observations on-the-ground at that PSU, be it from the GO observations or simply going out again to check the LC profile. The same procedure was used for labels tied for secondary classification. This was done for both the model and reference data. Using this PSU-level hierarchy, a comparison was performed on the

number of times each class appeared as a primary or secondary label in the reference and WorldCover datasets. For each class, aggregate count agreement was calculated by dividing the smaller count by the larger count. This metric measures how similar the aggregate class frequencies were across the 36 PSUs; it does not indicate whether the same individual PSUs received matching labels. The resulting class counts and aggregate count-agreement percentages are shown in Table 3. Separately, the dominant primary classes matched within 29 of the 36 PSUs, producing 80.6% PSU-level dominant-class agreement.

Land Cover Type	Reference Primary	Reference Secondary	WorldCover Primary	WorldCover Secondary	Primary Count Agreement	Secondary Count Agreement
Tree cover	0	24	3	19	0.0%	79.2%
Grassland	4	6	2	2	50.0%	33.3%
Shrubland	0	0	0	0	N/A	N/A
Bare/sparse vegetation	0	0	0	0	N/A	N/A
Permanent water bodies	3	0	2	1	66.7%	0.0%
Built-up	29	3	29	3	100.0%	100.0%
Cropland	0	0	0	1	N/A	0.0%
Herbaceous wetland	0	0	0	0	N/A	N/A

Table 3. Comparison of primary and secondary PSU class counts between the Map User and WorldCover datasets. Aggregate class-count agreement was calculated as the smaller class count divided by the larger class count. NA indicates that the class was absent from both datasets.

4. Discussion

4.1. Thematic Agreement Evaluation

As seen in the confusion matrix, the overall accuracy of WC sits at 65.2% for all LC classifications across Fremont. Data given in Table 2 gives further data about the confusion matrix, listing out the UA and PA for each LC type. Here it can be seen that WC has the lowest PA, equivalent to recall, when classifying SSUs with the label “Shrubland” and “Bare/sparse vegetation”. While WC data found there to be no instances of the “Shrubland” LC class at any of the SSUs, the reference data found there to be 105 instances of “Shrubland” labeled SSUs in the Fremont AOI. For “Bare/sparse vegetation”, while WC did identify 53 SSUs with this LC class, none of those SSUs lined up with the reference data’s “Bare/sparse vegetation” SSUs.

The opposite pattern occurred for the 28 SSUs classified as “Cropland” by WC, while the reference data found there to be no such SSUs anywhere within the Fremont AOI. While these results cannot be treated as absolute proof, as will be highlighted in Section 4.3, they suggest that WC classification limitations and spatial resolution may have contributed to the observed disagreements.

4.1.1. Resolution Issues

The latter could be true since “Shrubland” labels primarily referred to individual patches of shrubs within a single SSU, so without higher resolution, capturing such a small element of LC becomes more difficult, and so WC may have difficulty representing these smaller LC features because its 10 m pixels assign a single dominant class to each location. The same thing could be the case for the “Cropland” classifications produced by WC, where it is misinterpreting individual SSUs as being “Cropland” instead of “Grassland”, “Shrubland”, etc due to the lack of high resolution data making the difference between the labels harder to define.

This is generally evidenced by additional data from Figure 4, which illustrates the divide between WC and reference data labels, across classes. These inaccuracies, like the ones above, may again partly reflect spatial-resolution limitations. What’s interesting, however, is that the WC model very closely matches reference data when it comes to “Permanent water bodies” and “Built-up” labels. This may be due to the fact that these features often have higher contrast to the surrounding LC than the contrast between grasslands and tree cover. This pattern is consistent with spatial resolution being a possible contributor, since Permanent water bodies and Built-up surfaces may form larger or more visually distinct features than smaller patches of vegetation. However, this still does not prove that resolution was the sole cause of the observed disagreements.

4.1.2. WC Misclassifications

Aside from the resolution issue, it is also possible that WC is just generally misclassifying LC at PSU-size scales. This possibility is supported by Table 3, which contains data on PSU-scale trends when it comes to primary and secondary LC classifications. In terms of primary PSU classifications, WC differed notably from the reference data when it came to “Tree cover” identifying three PSUs as being primarily of that class while reference data showed that no such PSU exhibited those characteristics. WC also identified two PSUs as being “Grassland” dominant out of the four that reference data found. Taking this with other primary class agreement data, it suggests that the issue with data disagreement cannot be necessarily isolated purely to fine-scale spatial resolution alone, but WC’s general LC identification and misclassifications may also be a contributing factor. This is because Table 3’s metric doesn’t rely on fine details for LC, per SSU, instead focusing on measuring the ability of the model to take a PSU and identify larger-scale trends in terms of class hierarchy, rather than measuring if every individual SSU label matches. As such, higher-resolution data are not required to calculate this PSU-level metric because it aggregates the existing WC classifications across each PSU. However, spatial resolution may still affect the individual labels being aggregated and, therefore, the resulting PSU class hierarchy. Therefore, the persistence of disagreement at the PSU scale suggests that WC classification behavior may be a contributing factor in addition to spatial resolution, although this analysis cannot fully separate these effects.

This interpretation into WC’s shortcomings is further backed by the errors seen in secondary PSU class data from Table 3, which displays how the reference data suggests that 6 of the 36 PSUs have a secondary label of “Grassland”, while the WC data says that only 2 PSUs with such secondary classifications exist. Again, this demonstrates how WC may be classifying LC incorrectly at PSU-level scales, thus incorrectly estimating certain types of class label dominance.

An interesting thing to note is that WC’s error trends at PSU-scales do not seem to follow for the “Built-up” class. For this primary/secondary class label, WC appeared to perform much better when it came to accuracy. Specifically, WC had 100.0% aggregate count agreement when it came to labeling PSUs with primary or secondary classifications of “Built-up”, meaning the “Built-up” labels appeared as a primary or secondary class the same number of times in both datasets. Again, since the metrics in Table 3 were meant to assess general ability to predict LC without relying too much on resolution, WC’s performance on “Built-up” dominant PSUs suggests that it is uniquely good at classifying “Built-up” LC. Taking this in conjunction with the SSU data from Table 2, it becomes clear that WC’s errors cannot be fully attributed to resolution given that it consistently outperforms when it comes to labeling “Built-up” regions, as compared to other classes. As a result, this just further illustrates how some of the fault does lie with WC’s tendency to misclassify. Or to be more precise: WC’s inability to accurately classify LC which isn’t “Built-up”.

4.2. Potential Solutions

As listed previously, WC has misclassified certain regions of Fremont when it comes to the type of LC, or how prevalent a certain class of LC is in a specific PSU. As mentioned in Section 4.1, these issues could be due to various reasons, but can be broadly grouped into two categories: WC classification issues and remote sensing data issues.

4.2.1. Improvements to WC

To improve the issues seen with WC's raw misclassification of LC, a potential solution could be to retrain the model over more diversified LC data so that it performs better when it comes to predicting the LC of a region. This seems to be a valid solution to this current issue given the patterns seen in the misclassifications. From the data, it seems that WC is excellent at classifying LC of the "Built-up" class. This works well in areas which are highly developed, but less so in regions with less "Built-up" regions, thus limiting WC's accuracy over a broader category of landscapes which includes classes like "Shrubland" or "Bare/sparse vegetation".

4.2.2. Improvements to Satellite Data

The other issue raised in the previous section was issues with remote sensing data and the troubles that it creates for LC classification. As alluded to earlier, the issue here could be fixed by increasing the resolution for collected measurements. This may allow for higher precision in the data collected, thereby allowing for WC to make more precise predictions about the LC of an area, even at smaller scales. At a larger scale, this could lead to better overall estimation of how much of a certain region's LC is of a certain type of class, potentially improving WC's score on metrics found in the class hierarchy data of Table 3.

4.3. Limitations & Error Analysis

The results gathered in this paper were collected using the methodologies outlined above, but do carry some sources of error when it comes to the accuracy or precision of the data. Some of this error came from limitations in the collection of data through GO, while other sources of error stem from limitations in the process of comparing reference data to WC predictions or simply the collection of reference data.

4.3.1. GO Limitations & Errors

In terms of limitations during the GO data collection stage, some PSUs did not have observations which overlaid exactly with the PSU's land area, or any observations at all. Geolocation errors in precision made it difficult to know whether GO observations were being taken within the PSU, when the intended observations were to be taken at the edge of a PSU. The reason that some observations were intentionally taken at the edge of some PSUs, and not in the center of all of them, was primarily due to residential or crowded areas in the center of most of the PSUs. To protect privacy and avoid going into private property, some of the SSUs had their observations taken at the edge of the PSU, which then limited data accuracy due to geolocation precision limitations.

Aside from incorrectly aligned GO observations, some PSUs had no observations at all. Out of the 36 PSUs given to investigate, only 34 of them had associated GO observations. This was due to the fact that some PSUs were based entirely in residential or private property zones, making it so that such locations could not be accessed. As a result, GO observations were not taken for those 2 excluded PSUs.

Outside of spatial limitations, temporal limitations also played a part in the investigation. Given that the time allotted for GLOBE Observations fell on the shorter end, rapid assessment was used to quickly obtain the bare minimum data per AOI. The result was that more numeric metrics, such as confidence scoring or digital biometry, as outlined in Pixels and Pupils: The Integrated Field Operations Manual, simply had to be skipped to finish most of the necessary observations.

4.3.2. WC Comparison Errors

Given that WC data was primarily compared to the reference data generated through mixing GO data with other LC imagery, in the cases where the full PSU could not be seen in the GO observation, there existed increased dependence on other sources of local LC imagery to base the Reference labels off of. As a result, limitations in PSU GO observations would have led to potential errors in the generation of a reference dataset since labels were then based off of imagery which could have been taken at different times and dates.

Furthermore, after creating the reference dataset, it was stated previously that the labeling classes used were from LCMS. As such, a crosswalk between LCMS' and WC's labeling classes had to be performed to ensure a proper comparison between classifications. The issue with this lies in how different classes were defined. For example, LCMS' LC class of "Water" was used to label any type of water found in the Reference data, including pools, lakes, etc. However, for WC, the "water" class is replaced by "Permanent Water Bodies", and manmade structures like pools fit into "Built up". As such, in this example, areas with LC classified as water under LCMS would have had potential classification errors when crosswalking data labels in bulk. This error was not unique to this specific crosswalk either, such a mismatch would have also occurred with how LCMS defines mixed LC elements vs how WC is more rigid with LC classes. For example, again, if an area was labeled as "Grass shrub mix" under LCMS, it was crosswalked to WC classes as "Shrubland", effectively losing information about local LC in the area, increasing imprecision with LC descriptions. The result of this is potential errors when comparing to WC's data, since imprecision in classifying LC would mean that the reference data itself is inaccurate, thereby creating error within the data analysis/comparison.

Finally, it is key to note that the WC data used was from 2020, meaning that some errors may be attributable to development in Fremont in the time between the collection of reference data and the WC remote sensing data.

5. Conclusions

This study used a citizen science-informed reference-labeling workflow to evaluate WorldCover 2020 within the Fremont AOI. With regard to the methodology, the workflow illustrates how citizen science observations can support locally informed reference labeling and comparison with global land-cover products.

Beyond citizen science, however, this paper demonstrates that as far as we have come in the field of remote sensing data analysis and LC science, there still remain issues which must be resolved to obtain more accurate results. As discussed above, WC has been shown to have flaws when classifying LC in Fremont. Specifically, WC achieved 65.2% overall accuracy at the SSU level, while dominant PSU classes agreed in 29 of 36 PSUs (80.6%), indicating stronger correspondence at broader spatial scales. These findings identify potential areas for improvement in both the WorldCover classification workflow and its supporting remote sensing data. Although the numerical findings are specific to the Fremont AOI, the broader workflow of combining global land-cover products with locally informed validation may be useful in other heterogeneous landscapes when identifying potential limitations to existing LC prediction tools. However, further validation would be required before generalizing these results to other regions. As a result, future work could test whether higher-resolution, temporally matched imagery and more diverse reference data improve agreement across a wider variety of AOIs.

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Data Availability Statement: Data and supporting analysis outputs are available in Zenodo (doi: 10.5281/zenodo.21753308). Maps throughout this study were created using ArcGIS® software by Esri. ArcGIS® is the intellectual property of Esri and is used herein under license. Copyright © Esri. All rights reserved. For more information, please visit https://www.esri.com. Reference labels were generated by the researchers using USDA National Agricultural Imagery Program imagery, while model labels were

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